# Keshav Mishra

**AI Engineer**

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## SUMMARY

AI Engineer with 2 years of experience building LLM applications in Python, including retrieval-augmented generation (RAG) pipelines on the Qdrant vector database and agent workflows with LangGraph. Prior background in software quality assurance. Currently pursuing an M.Tech in Mathematics and Computing at NIT Delhi, with strong foundations in data structures, algorithms, and systems, and hands-on experience building vector search and RAG systems from scratch.

## EXPERIENCE

**AI Engineer** | Jobsier (Remote) *Jan 2024 – Jan 2026*

- Built and maintained Retrieval-Augmented Generation (RAG) pipelines in Python, grounding LLM responses in domain documents through semantic retrieval.
- Used Qdrant as the vector database for embedding storage and similarity search over document chunks.
- Designed multi-step, stateful LLM workflows using LangGraph to orchestrate retrieval, reasoning, and response generation.
- Developed Python services and REST APIs to expose AI features to the product, and worked with MySQL for application data.
- Collaborated in an agile team through code reviews, debugging, and iterative delivery of AI features.

**Manual QA Tester** *Aug 2022 – Dec 2023*

- Planned and executed manual test cases across the SDLC/STLC, covering functional, regression, smoke, sanity, and UAT testing.
- Logged, tracked, and verified defects in Jira, writing clear bug reports with reproduction steps and severity.
- Ran regression cycles for each release in Agile sprints and worked closely with developers to confirm fixes.

## PROJECTS

**Your Own AI: Vector Database and RAG Engine from Scratch** *C++17, HNSW, Ollama, REST API*

- Built a vector database in C++ implementing HNSW, KD-Tree, and brute-force search with cosine, Euclidean, and Manhattan distance, with a built-in benchmark to compare algorithms side by side.
- Implemented an end-to-end RAG pipeline: documents are split into overlapping 250-word chunks, embedded as 768-dimensional vectors with nomic-embed-text, indexed in HNSW, and the top-k chunks are passed to a local llama3.2 model to answer questions.
- Exposed a REST API (insert, delete, search, benchmark, document Q&A) and a web UI with a PCA scatter plot that visualizes semantic clusters.

**UPI Without Internet: Offline Payments over a Mesh Network** *Java 17, Spring Boot, JPA, H2*

- Built a Spring Boot backend that settles offline payments relayed device to device through a simulated mesh until a node with connectivity uploads them.
- Secured packets with hybrid encryption (RSA-OAEP with AES-256-GCM) so untrusted relay nodes cannot read or tamper with payments, and added replay protection using signed timestamps and nonces.
- Guaranteed exactly-once settlement with SHA-256 ciphertext hashing, atomic compare-and-set deduplication, transactional debit/credit, and optimistic locking; verified with a concurrency test of three bridge nodes delivering the same packet at once.

**Packet Analyzer: Deep Packet Inspection Engine** *C++17, Multithreading, TCP/IP*

- Built a DPI engine that reads PCAP captures, parses Ethernet, IP, TCP, and UDP headers, and tracks connections by 5-tuple flow.
- Extracted TLS SNI and HTTP Host fields to classify applications such as YouTube and Facebook, with rule-based blocking by IP, application, or domain.
- Designed a multithreaded pipeline with load-balancer and fast-path threads using consistent hashing and thread-safe queues, so all packets of a flow are handled by the same thread.

## TECHNICAL SKILLS

**Programming Languages:** Python, C++, SQL, Java
**LLM and GenAI:** Retrieval-Augmented Generation (RAG), LangGraph, Qdrant, vector search (HNSW), embeddings, Ollama
**AI/ML Foundations:** Supervised Learning, Neural Networks, NLP Basics; exploring TensorFlow, PyTorch, Scikit-learn
**Back-End and Databases:** REST APIs, MySQL, Relational Database Design
**Core CS Concepts:** Data Structures and Algorithms, OOP, Operating Systems, DBMS, Computer Networks, Cloud Computing
**Testing and Agile:** SDLC/STLC, Agile/Scrum, Jira, Regression Testing, Test Case Design, Defect Tracking, Test Automation Concepts, Debugging, Code Reviews
**Tools:** Git, GitHub

## EDUCATION

**M.Tech, Mathematics and Computing** | NIT Delhi *2026 – Present*

**B.Tech, Computer Science and Engineering** | GGSIPU, Delhi *2018 – 2022*

## ACHIEVEMENTS AND EXTRACURRICULAR

- Solved 300+ problems on LeetCode and Codeforces across arrays, hashing, trees, graphs, and dynamic programming.
- Participated in internal hackathons, building full-stack prototypes under time constraints.
- Actively contribute to open-source and personal GitHub projects, with a strong interest in building proprietary bots, assets, and new IP around Deep Learning and NLP for real-world business problems.